

Abstract

- **A single stepsize.** We run plain, unprojected TD(0) with

$$\eta_t = \frac{1}{c \tau_{\min} \phi_{\infty}^2 \sqrt{t+1} \log(t+3)},$$

which is independent of the curvature parameter ω and is used for both convergence regimes.

- **Simultaneous high-probability rates.** For the Polyak-Ruppert average $\bar{\theta}_T$, the TD potential satisfies

$$f(\bar{\theta}_T) - f(\theta^*) = \tilde{\mathcal{O}}\left(R_{\max}^2 \min\left\{\frac{\tau_{\min}^2 \phi_{\infty}^4}{\omega T}, \frac{\tau_{\min} \phi_{\infty}^2}{\sqrt{T}}\right\}\right).$$

Thus the same run attains both the fast curvature-dependent rate and the curvature-free rate.

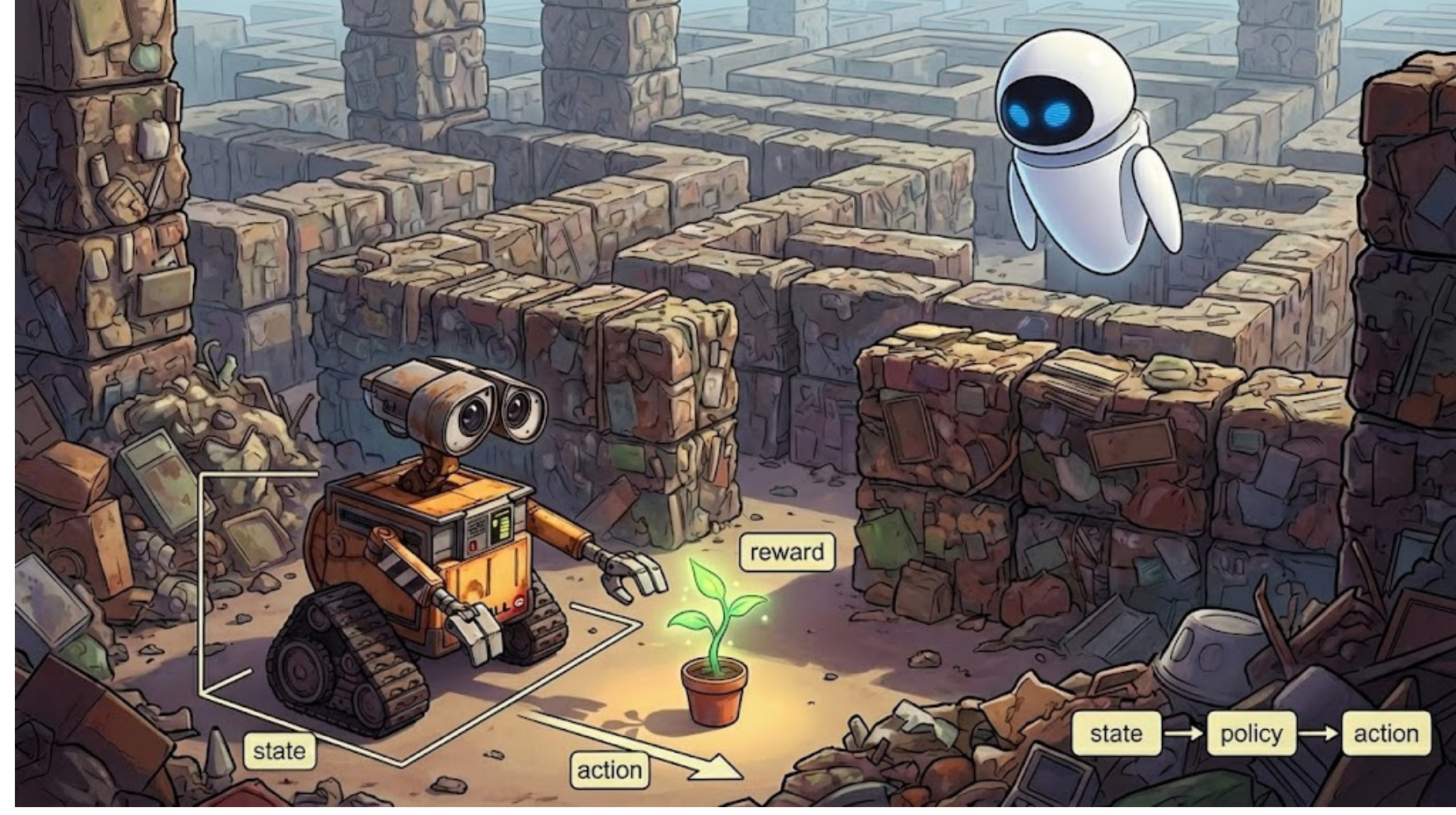
- **Implicit stability without projection.** With probability at least $1 - \delta$,

$$\sup_{t \geq 0} \|\theta_t\|_2 \leq R_{\max} = \rho(c, \delta) \max\left\{\|\theta_0 - \theta^*\|_2, \|\theta^*\|_2, \frac{r_{\infty}}{\phi_{\infty}}\right\},$$

where ρ is independent of ω . Hence bounded gradients are proved from the TD recursion rather than imposed by a projection radius.

Reinforcement Learning

- state space $\mathcal{S}$,
- action space $\mathcal{A}$,
- transition probability $P(s' | s, a)$,
- reward function $r(s, s')$,
- discount factor $\gamma \in [0, 1)$.



A **policy** μ specifies an action distribution $\mu(a | s)$. Starting from an initial state s_0 , the system evolves according to

$$a_t \sim \mu(\cdot | s_t), \quad s_{t+1} \sim P(\cdot | s_t, a_t),$$

and collects $r_t := r(s_t, s_{t+1})$ along one trajectory. Under irreducibility and aperiodicity, the state chain has transition matrix $\mathbf{P}^\mu$, where $\mathbf{P}^\mu(s, s') := \sum_{a \in \mathcal{A}} \mu(a | s) P(s' | s, a)$, and a unique stationary distribution $\mathbf{d}^\mu$. Its transition chain $\mathbf{Z}_t = (s_t, s_{t+1})$, with transition kernel P_Z and stationary law π_Z , satisfies, for some $\tau_{\min} \geq 1$, the **geometric mixing property**

$$\sup_z \left\| P_Z^k(z, \cdot) - \pi_Z \right\|_{\text{TV}} \leq 4 \cdot 2^{-k/\tau_{\min}}, \quad k \geq 0.$$

Temporal Difference Learning for Policy Evaluation

The goal of **policy evaluation** is to estimate

$$\mathbf{V}^\mu(s) := \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s, a_t \sim \mu(\cdot | s_t) \right].$$

For a large state space, use the linear approximation

$$\mathbf{V}_\theta(s) := \phi(s)^\top \theta, \quad \phi(s) \in \mathbb{R}^d.$$

Let $\Phi \in \mathbb{R}^{|\mathcal{S}| \times d}$ be the feature matrix, whose row for state s is $\phi(s)^\top$, so that $\mathbf{V}_\theta = \Phi \theta$. The TD target θ^* is the unique solution of

$$\Phi \theta^* = \Pi_D T^\mu(\Phi \theta^*), \quad T^\mu v := \mathbf{r}^\mu + \gamma \mathbf{P}^\mu v, \quad \mathbf{D} := \text{diag}(\mathbf{d}^\mu),$$

where $\mathbf{r}^\mu(s) := \sum_{s'} \mathbf{P}^\mu(s, s') r(s, s')$, and Π_D is the $\mathbf{D}$ -weighted projection onto $\text{span}(\Phi)$.

Given $\mathbf{Z}_t = (s_t, s_{t+1})$, the unprojected linear TD(0) update is

$$\theta_{t+1} = \theta_t + \eta_t \mathbf{g}(\theta_t, \mathbf{Z}_t) = \theta_t + \eta_t \delta_t \phi(s_t),$$

where $\delta_t = r_t + \gamma \phi(s_{t+1})^\top \theta_t - \phi(s_t)^\top \theta_t$ is the TD error.

Mean TD Dynamics and the Stability Bottleneck

Stationary mean dynamics. For $\mathbf{Z} = (s, s') \sim \pi_Z$, define

$$\bar{\mathbf{g}}(\theta) := \mathbb{E}_{\mathbf{Z} \sim \pi_Z} [\mathbf{g}(\theta, \mathbf{Z})].$$

Then θ^* is the TD fixed point:

$$\bar{\mathbf{g}}(\theta^*) = \mathbf{0}, \quad \langle \bar{\mathbf{g}}(\theta), \theta^* - \theta \rangle \geq \omega \|\theta - \theta^*\|_2^2,$$

where

$$\omega := (1 - \gamma) \lambda_{\min}(\Phi^\top \mathbf{D} \Phi) > 0.$$

Hence ω quantifies the conditioning of the mean TD dynamics and controls the fast rate.

Why the sample-path analysis is harder than i.i.d. SGD. The iterate θ_t is a function of the entire past $\mathbf{Z}_0, \dots, \mathbf{Z}_{t-1}$, while $\mathbf{Z}_t$ is generated from the same Markov chain. Consequently, in general, the current noise $\mathbf{g}(\theta_t, \mathbf{Z}_t) - \bar{\mathbf{g}}(\theta_t)$ is not conditionally mean-zero with respect to $\mathcal{F}_{t-1} := \sigma(\mathbf{Z}_0, \dots, \mathbf{Z}_{t-1})$:

$$\mathbb{E}[\mathbf{g}(\theta_t, \mathbf{Z}_t) | \mathcal{F}_{t-1}] \neq \bar{\mathbf{g}}(\theta_t).$$

The stability feedback loop. For every θ and state pair $\mathbf{Z}$,

$$\|\mathbf{g}(\theta, \mathbf{Z})\|_2 \leq r_{\infty} \phi_{\infty} + 2\phi_{\infty}^2 \|\theta\|_2.$$

A loose bound on $\|\theta_t\|_2$ makes the update bound loose, which feeds into θ_{t+1} and can recursively explode. Projection-based proofs break this loop by imposing a radius.

Main Results

Theorem: Single-stepsize unprojected TD(0)

Assume a finite, irreducible, aperiodic Markov reward process (MRP) with a full-column-rank feature matrix Φ , $\|\phi(s)\|_2 \leq \phi_{\infty}$, and $|r(s, s')| \leq r_{\infty}$. For any $\delta \in (0, 1)$ and sufficiently large $c = c(\delta)$, run

$$\eta_t = \frac{1}{c \tau_{\min} \phi_{\infty}^2 \sqrt{t+1} \log(t+3)}.$$

Let

$$R_{\text{base}} := \max\left\{\|\theta_0 - \theta^*\|_2, \|\theta^*\|_2, \frac{r_{\infty}}{\phi_{\infty}}\right\}, \quad R_{\max} := \rho R_{\text{base}}, \quad S_T := \sum_{t=0}^{T-1} \eta_t, \quad \bar{\theta}_T := \frac{1}{S_T} \sum_{t=0}^{T-1} \eta_t \theta_t.$$

Here $\rho > 2$ is a fixed constant dependent of only c and δ . For the TD potential

$$f(\theta) = (1 - \gamma) \|\mathbf{V}_\theta - \mathbf{V}_{\theta^*}\|_{\mathbf{D}}^2 + \gamma \|\mathbf{V}_\theta - \mathbf{V}_{\theta^*}\|_{\text{Dir}}^2,$$

where

$$\|v\|_{\mathbf{D}}^2 = v^\top \mathbf{D} v, \quad \|v\|_{\text{Dir}}^2 := \frac{1}{2} \sum_{s, s'} \mathbf{d}^\mu(s) \mathbf{P}^\mu(s, s') (v(s) - v(s'))^2.$$

With probability at least $1 - \delta$, simultaneously for every $T \geq 1$,

$$\sup_{t \geq 0} \|\theta_t\|_2 \leq R_{\max},$$

and

$$f(\bar{\theta}_T) - f(\theta^*) \leq \tilde{\mathcal{O}}\left(R_{\max}^2 \min\left\{\frac{\tau_{\min}^2 \phi_{\infty}^4}{\omega T}, \frac{\tau_{\min} \phi_{\infty}^2}{\sqrt{T}}\right\}\right).$$

Comparisons

Rate	Paper	Algorithm Design		Guarantees	
		Extra inputs	No projection	Optimal dependence on ω	High probability
$\tilde{\mathcal{O}}(1/T)$	Bhandari et al. [2018]	ω	$\times$	$\checkmark$	$\times$
	Srikant and Ying [2019]	$\omega, \ \theta^*\ _2$	$\checkmark$	$\checkmark$	$\times$
	Patil et al. [2023]	—	$\checkmark$	$\checkmark$	$\times$
	Samsonov et al. [2024]	—	$\checkmark$	$\checkmark$	$\checkmark$
	Mitra [2024]	ω	$\checkmark$	$\checkmark$	$\times$
	Li et al. [2025]	ω	$\checkmark$	$\checkmark$	$\times$
$\tilde{\mathcal{O}}(1/\sqrt{T})$	Our work	—	$\checkmark$	$\checkmark$	$\checkmark$
	Bhandari et al. [2018]	—	$\times$	$\checkmark$	$\times$
	Liu and Olshevsky [2021]	—	$\times$	$\checkmark$	$\times$
	Sun et al. [2021]	—	$\times$	$\times$	$\times$
	Our work	—	$\checkmark$	$\checkmark$	$\checkmark$

Proof Idea: A Poissonized Self-Bounding Bootstrap

1. Induct on the past; use $\theta_t - \theta^*$ as the surrogate. Let

$$\mathbf{e}_t := \theta_t - \theta^*,$$

and assume only

$$\text{IH}_{t-1} : \max_{0 \leq i \leq t-1} \|\theta_i\|_2 \leq R_{\max}.$$

Expanding around the fixed point yields

$$\begin{aligned} \|\mathbf{e}_t\|_2^2 &= \|\mathbf{e}_0\|_2^2 + \sum_{i=0}^{t-1} \eta_i^2 \|\mathbf{g}(\theta_i, \mathbf{Z}_i)\|_2^2 + 2 \sum_{i=0}^{t-1} \eta_i \langle \bar{\mathbf{g}}(\theta_i), \theta_i - \theta^* \rangle + 2 \sum_{i=0}^{t-1} \eta_i h_i(\mathbf{Z}_i) \\ &\leq \|\mathbf{e}_0\|_2^2 + Q_t + 2B_t, \\ Q_t &:= \sum_{i=0}^{t-1} \eta_i^2 \|\mathbf{g}(\theta_i, \mathbf{Z}_i)\|_2^2, \quad B_t := \sum_{i=0}^{t-1} \eta_i h_i(\mathbf{Z}_i), \end{aligned}$$

where

$$h_i(z) := \langle \mathbf{g}(\theta_i, z) - \bar{\mathbf{g}}(\theta_i), \theta_i - \theta^* \rangle,$$

and

$$\langle \bar{\mathbf{g}}(\theta_i), \theta_i - \theta^* \rangle \leq 0.$$

Every iterate-dependent term on the right uses only θ_i with $i \leq t - 1$. Under IH_{t-1} ,

$$\|\mathbf{g}(\theta_i, z)\|_2 \leq 3\phi_{\infty}^2 R_{\max}, \quad \|\mathbf{e}_i\|_2 \leq 2R_{\max}, \quad \sum_{i \geq 0} \eta_i^2 < \infty.$$

Thus, if B_t is uniformly controlled, the bound on the new iterate follows from $\|\theta_t\|_2 \leq \|\mathbf{e}_t\|_2 + \|\theta^*\|_2$.

2. Poissonize the only hard term B_t . Freeze the past iterate θ_i . Since $\mathbb{E}_{\mathbf{Z} \sim \pi_Z} [h_i(\mathbf{Z})] = 0$, define

$$(P_Z u)(z) := \mathbb{E}[u(\mathbf{Z}_1) | \mathbf{Z}_0 = z], \quad u_i - P_Z u_i = h_i, \quad u_i(z) = \sum_{\ell=0}^{\infty} (P_Z^\ell h_i)(z).$$

Geometric mixing gives

$$\|u_i\|_{\infty} \leq 16\tau_{\min} \|h_i\|_{\infty}.$$

The Poisson identity decomposes one Markov-noise term as

$$h_i(\mathbf{Z}_i) = \underbrace{u_i(\mathbf{Z}_{i+1}) - (P_Z u_i)(\mathbf{Z}_i)}_{\Delta M_{i+1}} + \underbrace{u_i(\mathbf{Z}_i) - u_i(\mathbf{Z}_{i+1})}_{\text{boundary remainder}}.$$

Moreover, IH_{t-1} implies

$$\|h_i\|_{\infty} \lesssim \phi_{\infty}^2 R_{\max}^2 \implies \|u_i\|_{\infty} \lesssim \tau_{\min} \phi_{\infty}^2 R_{\max}^2.$$

Hence

$$|\eta_i \Delta M_{i+1}| \lesssim \frac{R_{\max}^2}{c \sqrt{i+1} \log(i+3)}, \quad \sum_{i \geq 0} \|\eta_i \Delta M_{i+1}\|_{\infty}^2 < \infty.$$

With $M_t := \sum_{i=0}^{t-1} \eta_i \Delta M_{i+1}$, Pinelis' inequality therefore controls $\sup_t |M_t|$ with high probability.

3. Summation by parts makes the remainder past-only. Write $B_t = M_t + R_t$. An equivalent Abel decomposition is

$$\begin{aligned} R_t &= \eta_0 u_0(\mathbf{Z}_0) - \eta_{t-1} u_{t-1}(\mathbf{Z}_t) - \sum_{i=1}^{t-1} (\eta_{i-1} - \eta_i) u_{i-1}(\mathbf{Z}_i) \\ &\quad + \sum_{i=1}^{t-1} \eta_i (u_i - u_{i-1})(\mathbf{Z}_i). \end{aligned}$$

The first three terms use bounded u_i , while the last term uses local Lipschitz continuity together with

$$\theta_i - \theta_{i-1} = \eta_{i-1} \mathbf{g}(\theta_{i-1}, \mathbf{Z}_{i-1}),$$

which gives

$$\|u_i - u_{i-1}\|_{\infty} \lesssim \tau_{\min} \phi_{\infty}^4 R_{\max}^2 \eta_{i-1}.$$

Thus, $R_t = \mathcal{O}(\tau_{\min} R_{\max}^2 / c + R_{\max}^2 / c^2)$, which is small for large c .

4. Close the self-bounding bootstrap. Combining the quadratic, martingale, and remainder bounds gives, with high probability, for some finite constants $A_1(\delta) > 0$ and $A_2 > 0$,

$$\|\theta_t\|_2^2 \leq \left(4 + \frac{A_1(\delta) \rho^2}{c} + \frac{A_2 \rho^2}{c^2}\right) R_{\text{base}}^2 \leq \rho^2 R_{\text{base}}^2 = R_{\max}^2,$$

for a sufficiently large stepsize multiplier c . Hence $\text{IH}_{t-1} \Rightarrow \text{IH}_t$.